Domino Startup Concept

Problem: Domino transplantation could save 2,000+ additional lives annually in the US, but only 50-100 domino transplants happen because:

1. Transplant coordinators lack tools to identify optimal Patient C candidates from waitlists (manual screening takes 8-12 hours per domino opportunity)
2. Risk-benefit calculations are subjective and inconsistent across hospitals
3. Complex multi-patient logistics cause 30-40% of potential domino chains to fail
4. Informed consent is difficult - patients can't visualize long-term risks vs. immediate survival benefit

Solution: AI-powered platform that identifies optimal domino recipients, calculates personalized survival probabilities, coordinates multi-patient logistics, and generates patient-friendly decision support materials.

Market Opportunity:

- 250+ transplant centers in US
- 30,000+ patients on liver transplant waitlist
- 500-800 FAP/metabolic disease patients transplanted annually (potential domino donors)
- Current domino utilization rate: 10-15%
- Target domino utilization rate: 40-50% = 1,500 additional lives saved/year

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What You Build in 11 Hours

Core Features:

1. Domino Candidate Matcher (3 hours)

- Input: Upload Patient B details (metabolic disease type, liver function) + Patient C waitlist data (CSV)
- AI Analysis:
    - Scores each waitlist patient for domino suitability based on:
        - Age (60+ preferred)

        - MELD score (urgency)
        - Life expectancy without transplant (<6 months optimal)
        - Expected lifespan with domino liver (>10 years valuable)
        - Comorbidities that might kill them before metabolic disease manifests
    - Ranks top 10 candidates with explanations

2. Personalized Survival Calculator (3 hours)

- Interactive Tool:
    - Patient C enters their data (age, diagnosis, current health status)
    - Two scenarios displayed side-by-side:
        - Option A: Wait for standard donor (survival probability curve)
        - Option B: Accept domino liver (survival probability curve with metabolic disease onset marked)
    - Visual timeline showing: "You gain 12 years of life expectancy vs. waiting"
    - Trade-off visualization: "90% chance you survive to see grandchildren graduate vs. 30% chance waiting for standard donor"

3. Informed Consent Generator (2 hours)

- AI-Powered Document Creation:
    - Generates patient-friendly consent forms explaining:
        - What domino transplant is (layman's terms with diagrams)
        - Specific risks for their metabolic disease type
        - Personalized timeline of when symptoms might appear
        - Comparison to "do nothing" scenario
    - Available in multiple languages
    - Includes visual infographics

4. Domino Chain Coordinator Dashboard (2 hours)

- Timeline Management:

- Tracks all three parties: Donor A → Patient B → Patient C
- Synchronizes surgical schedules across teams
- Alerts for timing conflicts
- Organ transport logistics
- Color-coded status tracker (Green: on schedule, Red: delay risk)

5. Bonus Feature: Outcome Predictor (1 hour if time permits)

- Upload historical domino transplant data
- ML model predicting success probability based on patient characteristics
- "Patients like you have 85% 10-year survival rate"

11-Hour Build Plan

Hour 0-1: Setup & Data

- Set up Next.js + Tailwind project
- Find/create sample datasets:
  - Mock waitlist data (50 patients with MELD scores, ages, diagnoses)
  - FAP disease progression data (from medical literature)
  - Survival curve data for liver failure patients
- Set up OpenAI API access

Hour 1-3: Matching Algorithm

Frontend:

- Upload interface for Patient B profile + waitlist CSV
- Results table showing top 10 matches

Backend:

- Scoring algorithm considering:
  - Age weight (60-75 = high score)
  - MELD score (higher = more urgent = higher score)

    - Comorbidity risk (diabetes, heart disease = higher score - likely to die before metabolic disease)
    - Blood type compatibility
- GPT-4 generating natural language explanations: "John Smith (68) is optimal because..."

Hour 3-6: Survival Calculator

Frontend:

- Clean, interactive interface with dual survival curves
- Slider inputs for patient parameters
- Real-time graph updates
- Export to PDF button

Backend/Logic:

- Mathematical models for survival probability:
    - Standard waitlist survival (exponential decay based on MELD)
    - Domino liver survival (plateau for 10-15 years, then decline)
- Chart.js for beautiful visualizations
- Calculate key metrics:
    - "Expected additional years of life"
    - "Probability of surviving to age X"
    - "Risk of metabolic symptoms by year"

Hour 6-8: Consent Generator + Dashboard

Consent Generator:

- Template-based generation with patient-specific data insertion
- GPT-4 creating patient-appropriate explanations
- Visual timeline generator
- PDF export with diagrams

Dashboard:

- Kanban-style board showing domino chain progress
- Three columns: Patient A (donor), Patient B (recipient/donor), Patient C (final recipient)
- Timeline view with surgical scheduling
- Alert system for conflicts

Hour 8-9: Demo Preparation

- Create 3 realistic patient scenarios
- Record demo video walkthrough
- Screenshot key features
- Prepare sample outputs (PDFs, reports)

Hour 9-10: Pitch Deck

- Problem slide (with statistics on wasted domino opportunities)
- Solution demo (screenshots + live demo plan)
- Market size
- Business model
- Traction plan (pilot with 3 transplant centers)

Hour 10-11: Polish & Practice

- Fix bugs
- Improve UI/UX
- Practice 5-minute pitch
- Prepare for Q&A

Tech Stack (All Feasible in 11 Hours)

Frontend:

- Next.js + React
- Tailwind CSS for rapid UI

- Chart.js or Recharts for survival curves
- React PDF for document generation

Backend/AI:

- OpenAI GPT-4 API for matching explanations and consent generation
- Simple scoring algorithms (no complex ML training needed)
- Firebase or Supabase for quick database (optional)

Data:

- Mock datasets (CSV files)
- Public medical literature for disease progression rates
- UNOS (organ network) publicly available statistics

Demo Strategy for Judges

Opening Hook (30 seconds):

"Every year, 400 people donate organs that could save TWO lives instead of one. But we only capture 50 of these opportunities because hospitals can't efficiently match domino recipients. DominoMatch uses AI to turn one donor into two survivors."

Live Demo (3 minutes):

Scenario 1 - The Matching:

1. "This is Sarah, 42, with FAP. She's getting a new liver today."
2. Upload Sarah's profile + waitlist of 50 patients
3. AI instantly identifies top 10 candidates
4. Click on Michael, 67, with acute liver failure
5. Show explanation: "Michael is optimal because he's 67 (limited life expectancy), MELD score 32 (will die in 2 weeks without transplant), and has diabetes (likely to die before FAP symptoms emerge in 15 years)"

Scenario 2 - The Decision Support:

1. Michael's family wants to understand the trade-off

2. Open Survival Calculator
3. Show two curves side-by-side:
    - Red curve: "Wait for standard donor" → 15% chance of surviving 6 months
    - Green curve: "Accept domino liver" → 85% chance of surviving 10+ years
4. Highlight: "Michael gains 9.4 expected years of life"
5. Generate informed consent PDF in real-time

Scenario 3 - The Coordination:

1. Show dashboard tracking all three patients
2. Demonstrate scheduling conflict detection
3. "Surgery must happen within 8-hour cold ischemia time—our platform ensures perfect coordination"

Closing Impact Statement:

"DominoMatch could increase domino transplant utilization from 10% to 50%, saving an additional 1,500 lives annually in the US alone. At $200/month per transplant center, that's a $600K ARR market with lives saved as our North Star metric."

Business Model

Revenue Streams:

SaaS Model:

- $200/month per transplant center (250 centers in US = $600K ARR potential)
- Enterprise tier: $500/month (includes API access, custom integrations)

Per-Transaction Model (Alternative):

- $500 per successful domino match facilitated
- Aligns incentives with outcomes

Data Licensing:

- Anonymized domino transplant outcome data to researchers
- $50K-100K annually

Go-to-Market:

Phase 1 (Months 1-6): Pilot with 3 high-volume transplant centers

- Focus on FAP domino liver transplants (easiest use case)
- Prove 3x increase in domino utilization

Phase 2 (Months 6-12): Expand to 20 centers

- Add heart-lung domino capabilities
- Build integrations with UNOS (national organ network)

Phase 3 (Year 2): National rollout

- Partner with transplant societies
- Publish clinical outcomes data
- Expand internationally (Europe has high domino adoption)

Competitive Advantages

1. First-mover in domino-specific software - No existing specialized tools
2. AI-powered matching - Faster and more objective than manual review
3. Patient-facing decision support - Improves informed consent process
4. Lives-saved metric - Measurable impact attracts funding/partnerships
5. Network effects - More centers = better data = better matching algorithms

BioInventory AI - Lab Supply Expiration & Usage Tracker

Problem: Research labs waste $50K-200K annually on expired reagents because tracking 300+ items manually is impossible. Running out of critical reagents mid-experiment delays projects by weeks. No visibility into "when should I reorder?"

Solution: Photo-based inventory system where researchers photograph reagent labels, AI extracts product name/lot number/expiration date, tracks usage patterns, predicts reorder timing, and sends expiration alerts.

What You Build (11 hours):

- Mobile-first web app with camera interface
- GPT-4 Vision extracting text from reagent bottle labels (even handwritten)
- Inventory database with expiration tracking
- Usage prediction: "You use this reagent every 3 weeks, you have 1 bottle left, order now"
- Alert system: "3 items expiring in next 14 days"
- Shopping list generator grouped by vendor

Demo Strategy:

- Photograph 5 different reagent bottles with varying label styles
- Show AI extracting all data correctly (including hard-to-read handwriting)
- Display dashboard with color-coded expiration timeline
- Show predictive reorder alert: "Order Tris buffer by Nov 20 based on usage pattern"

Unique Value: First system using computer vision for reagent label reading—current inventory tools require manual barcode scanning or data entry. Works with ANY label format.

Target: Research labs, biotech companies, hospital labs, pharmaceutical QC facilities

Tech Stack: Next.js + GPT-4 Vision API + Firebase for database + Calendar/alert logic + mobile-optimized UI